past paper 2022

(Short Questions)

Q1 (i)

Solve the given differential equation.

$$x\frac{dy}{dx} = y + \sqrt{x^2 - y^2} \quad ; \ y(x_0) = 0 \ \text{ where } x_0 > 0$$

soln:

$$x\frac{dy}{dx} = y + \sqrt{x^2 - y^2}$$

$$\frac{dy}{dx} = \frac{y + \sqrt{x^2 - y^2}}{x} \quad \text{---(i)}$$

put

$$y = vx$$

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

put in (i)

$$v + x\frac{dv}{dx} = \frac{vx + \sqrt{x^2 - v^2x^2}}{x}$$

$$v + x\frac{dv}{dx} = \frac{x\left(v + \sqrt{1 - v^2}\right)}{x}$$

$$x\frac{dv}{dx} = v + \sqrt{1 - v^2} - v$$

$$\frac{dv}{\sqrt{1-v^2}} = \frac{dx}{x}$$

Integrate both sides

$$\sin^{-1}(v) = \ln x + C$$

put $v = y/x$

$$\sin^{-1}(y/x) - \ln x = C \quad ---(ii)$$

For condition

$$y(x_0) = 0 \quad , \text{ when } x = x_0 , y = 0$$

put in (ii)

$$\sin^{-1}(0/x_0) - \ln x_0 = C$$

$$C = -\ln x_0$$

so, $\sin^{-1}(y/x) - \ln x = -\ln x_0$

$$\sin^{-1}(y/x) + \ln x_0 - \ln x = 0$$

$$\sin^{-1}(y/x) + \ln\left(\frac{x_0}{x}\right) = 0 \quad \textbf{(Ans.)}$$

Q12 (ii)

Solve intial value problem

$$\mathbf{(\cos x \sin x - xy^2)dx + y(1-x^2)dy = 0}$$

$$\mathbf{y(0) = 2}$$

soln:

$$(\cos x \sin x - xy^2)dx + y(1-x^2)dy = 0$$

$$M = \cos x \sin x - xy^2$$

$$N = y - x^2 y$$

$M_y = -2xy$, $N_x = -2xy$

so, $M_y = N_x$

so, D.Eq is Exact

therefore solution is

$$\int M dx + \int [\text{Term of } N \text{ free of } x] dy = C$$

$$\int (\cos x \sin x - xy^2) dx + \int y\, dy = C$$

$$-\int \cos x(-\sin x) dx - y^2 \int x\, dx + \int y\, dy = C$$

$$-\left[\frac{\cos^2 x}{2}\right] - \frac{x^2 y^2}{2} + \frac{y^2}{2} = C$$

$$y^2 - x^2 y^2 - \cos^2 x = 2C$$

$$y^2(1-x^2) - \cos^2 x = C_1 \text{ --- (i)}$$

For condition

$y(0) = 2$; when $x = 0, y = 2$

put in (i)

$$(2)^2(1-0) - \cos^2(0) = C_1$$

$$4 - 1 = C_1 \Rightarrow C_1 = 3$$

so, $y^2(1-x^2) - \cos^2 x = 3$

$$y^2(1-x^2) - \cos^2 x - 3 = 0 \quad \text{(Ans.)}.$$

Q1 (iii)

Define Regular singular point.

Ans: Let

$$y'' + p(x)y' + Q(x)y = 0 \text{ --- (i)}$$

be a 2nd order linear

homogenous O.D.Eq and

$x = x_0$ is a singular point of eq

i.e ($p(x)$ or $Q(x)$ is not analytical at $x = x_0$)

and $(x-x_0)p(x)$ and $(x-x_0)^2 Q(x)$ are analytical at $x = x_0$

then $x = x_0$ is regular singular point.

Note :- $(x-x_0)\,p(x)$ is analytic iff $\lim_{x \to x_0} (x-x_0)\,p(x)$ exists

$(x-x_0)^2 Q(x)$ is analytical iff $\lim_{x \to x_0} (x-x_0)^2 Q(x)$ exists

Find regular singular point of $xy'' - 2xy' + 5y = 0$

soln: $y'' - 2y' + \frac{5}{x} y = 0$

$p(x) = -2$

$Q(x) = 5/x$

$x = 0$ is a singular point.

$\lim_{x \to 0} (x-0)\,p(x) = \lim_{x \to 0} x(-2) = 0$

$\lim_{x \to 0} (x-0)^2 Q(x) = \lim_{x \to 0} x^{\cancel{2}}\left(\frac{5}{\cancel{x}}\right) = 0$

$(x-0)$ is regular singular point

Q1 (iv)

Find power series solution about ordinary point $x=0$

$$(x-1)y''+y'=0$$

soln: let $y = \sum_{n=0}^{\infty} C_n x^n$

$$y' = \sum_{n=1}^{\infty} n C_n x^{n-1}, \quad y'' = \sum_{n=2}^{\infty} n(n-1) C_n x^{n-2}$$

so, $(x-1)\sum_{n=2}^{\infty} n(n-1) C_n x^{n-2} + \sum_{n=1}^{\infty} n C_n x^{n-1} = 0$

$$\sum_{n=2}^{\infty} n(n-1) C_n x^{n-1} - \sum_{n=2}^{\infty} n(n-1) C_n x^{n-2} + \sum_{n=1}^{\infty} n C_n x^{n-1} = 0$$

$$\sum_{n=1}^{\infty} n(n+1) C_{n+1} x^n - \sum_{n=0}^{\infty} (n+2)(n+1) C_{n+2} x^n + \sum_{n=0}^{\infty} (n+1) C_{n+1} x^n = 0$$

$$\sum_{n=1}^{\infty} n(n+1) C_{n+1} x^n - 2.1 C_2 x^0 - \sum_{n=1}^{\infty} (n+1)(n+2) C_{n+2} x^n$$

$$+ C_1 x^0 + \sum_{n=1}^{\infty} (n+1) C_{n+1} x^n = 0$$

$$\sum_{n=1}^{\infty} \{(n+1) C_{n+1} + n(n+1) C_{n+1} - (n+1)(n+2) C_{n+2}\} x^n$$

$$+ (C_1 - 2C_2) x^0 = 0$$

Equating coefficient of like powers

$$(n(n+1)+(n+1)) C_{n+1} = (n+1)(n+2) C_{n+2}$$

$$C_{n+2} = \frac{(n+1)(n+1)\, C_{n+1}}{(n+1)(n+2)}$$

$$C_{n+2} = \frac{(n+1)}{(n+2)} C_{n+1} \quad \text{--- (i)} \quad n = 1, 2,$$

$$C_1 - 2C_2 = 0$$

$$C_2 = \frac{C_1}{2} \quad \text{--- (ii)}$$

from (i)

$$C_3 = \frac{2}{3} C_2 \Rightarrow \frac{2}{3}\frac{C_1}{2} \Rightarrow \frac{C_1}{3}$$

$$C_4 = \frac{3}{4} C_3 \Rightarrow \frac{C_1}{4}$$

$$C_5 = \frac{4}{5} C_4 \Rightarrow \frac{C_1}{5}$$

$$C_6 = \frac{5}{6} C_5 \Rightarrow \frac{C_1}{6}$$

The series soln is

$$y = \sum_{n=0}^{\infty} C_n x^n$$

$$y = C_0 x^0 + C_1 x + C_2 x^2 + C_3 x^3 + \cdots$$

$$y = C_0 x^0 + C_1 x + \frac{C_1}{2} x^2 + \frac{C_1}{3} x^3 + \frac{C_1}{4} x$$

$$y = C_0 x^0 + C_1 x\left(1 + \frac{x}{2} + \frac{x^2}{3} + \frac{x^3}{4} + \cdots\right.$$

Q1 (v)

verify that piecewise function below is a soln of given D

$$xy' - 2y = 0 \; ; \; y = \begin{cases} -x^2, & x < 0 \\ x^2, & x > 0 \end{cases}$$

soln: If $x<0$

$y=-x^2$

$y'=-2x$

put in eq (i)

L.H.S

$= x(-2x)-2(-x^2)$

$= -2x^2+2x^2$

$= 0 = $ R.H.S

If $x>0$

$y=x^2$

$y'=2x$

L.H.S

$= x(2x)-2(x^2)$

$= 2x^2-2x^2$

$= 0 = $ R.H.S

so, $(-x^2, x^2)$ is the solution of the equation

Q1.(vi)

Determine whether given equation is Exact. If a otherwise solve.

$(\sin y - y\sin x)dx + (\cos x + x\cos y - y)dy = 0$

soln:

$M = \sin y - y\sin x$
$N = \cos x + x\cos y - y$

$M_y = \cos y - \sin x$
$N_x = -\sin x + \cos y$
$M_y = N_x$
so D. equation is exact
Now soln is
$\int M dx + \int [\text{term of } N \text{ free of } x] dy =$
$\int (\sin y - y\sin x) dx + \int -y \, dy = C$

$x\sin y - y(-\cos x) - \frac{y^2}{2} = C$

$x\sin y + y\cos x - \frac{y^2}{2} = C$

$2x\sin y + 2y\cos x - y^2 = 2C$
$2x\sin y + 2y\cos x - y^2 = C_1$. (Ans.

(Long Qestions)

Q2

Solve the differential Eq subject to intial condition.
$(x+y)^2 dx + (2xy + x^2 - 1) dy = 0$

soln:

$(x+y)^2 dx + (2xy + x^2 - 1) dy = 0$ --- (i)

$M = (x+y)^2$

$N = (2xy + x^2 - 1)$

$M_y = 2(x+y).1$

$M_y = 2(x+y)$

$N_x = 2y + 2x$

$M_y = N_x$

so D.Eq is exact

Now solution is

$\int M dx + \int (\text{Term of } N \text{ free of } x) dy = C$

$\int (x+y)^2 dx + \int -1 \, dy = C$

$\int (x^2 + 2xy + y^2) dx - y = C$

$\frac{x^3}{3} + x^2 y + xy^2 - y = C$

$x^3 + 3x^2 y + 3xy^2 - 3y = 3C$

$x^3 + 3xy(x+y) - 3y = C_1$ --- (ii)

For condition

$y(1) = 1$; when $x = 1, y = 1$

$(1)^3 + 3(1)(1)(1+1) - 3(1) = C_1$

$1 + 6 - 3 = C_1$

$C_1 = 4$

put in (ii)

$x^3 + 3xy(x+y) - 3y - 4 = 0$ **(Ans)**

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Q6

Solve the following D.Eq

$y'' - 2y' + 2y = e^x \tan x$

soln:

$(D^2 - 2D + 2)y = e^x \tan x$

For yc

$(D^2 - 2D + 2)y = 0$

Auxilary eq is

$m^2 - 2m + 2 = 0$

$m = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(2)}}{2(1)}$

$= \frac{2 \pm \sqrt{4-8}}{2}$

$m = \frac{2 \pm 2i}{2} \Rightarrow 1 \pm i$

$yc = e^x (A\cos x + B\sin x)$

$yc = e^x A\cos x + e^x B\sin x$

$y_1 = e^x \cos x, \ y_2 = e^x \sin x$

$y_1' = e^x \cos x - e^x \sin x, \ y_2' = e^x \sin x + e^x \cos x$

$W = y_1 y_2' - y_2 y_1'$

$W = e^x \cos x (e^x \sin x + e^x \cos x) - e^x \sin x (e^x \cos x - e^x \sin x)$

$= e^{2x} \cos x \sin x + e^{2x} \cos^2 x - e^{2x} \cos x \sin x + e^{2x} \sin^2 x$

$$= e^{2x}(\sin^2 x + \cos^2 x)$$

$$W = e^{2x}$$

$$y_p = u_1 y_1 + u_2 y_2$$

$$F(x) = e^x \tan x$$

$$u_1 = \int \frac{-y_2 F(x)}{W} dx$$

$$= \int \frac{-e^x \sin x \, e^x \tan x}{e^{2x}} dx$$

$$u_1 = -\int \frac{\sin^2 x}{\cos x} dx$$

put $\sin x = t$

$\cos x \, dx = dt \Rightarrow dx = \dfrac{dt}{\cos x}$

$$u_1 = -\int \frac{t^2}{\cos^2 x} dt$$

$$= -\int \frac{t^2}{1-t^2} dt$$

$$= \int \frac{-t^2+1-1}{1-t^2} dt$$

$$= \int 1\, dt - \int \frac{1}{1-t^2} dt$$

$$= t - \frac{1}{2} \ln\left|\frac{1+t}{1-t}\right|$$

$$u_1 = \sin x - \frac{1}{2} \ln\left|\frac{1+\sin x}{1-\sin x}\right|$$

$$u_2 = \int \frac{y_1 f(x)}{W} dx$$

$$= \int \frac{e^x \cos x \; e^{x} \tan x}{e^{2x}} dx$$

$$= \int \sin x \, dx$$

$$u_2 = -\cos x$$

so,

$$y_p = e^x \cos x \left[\sin x - \frac{1}{2} \ln \left| \frac{1+\sin x}{1-\sin x} \right| \right] + e^x \sin x (-\cos x)$$

$$y_p = -e^x \cos x \left[\frac{1}{2} \ln \left| \frac{1+\sin x}{1-\sin x} \right| \right]$$

Q3

Check whether the given functio are linearly dependent or not $f_1(x) = \cos 2x$, $f_2(x) = 1$, $f_3(x) = \cos^2 x$

soln: let $a(1) + b\cos 2x + C\cos^2 x = 0$

put $x = 0$; $a + b + C = 0$ --- (i)

put $x = \pi/2$; $a + b = 0$ --- (ii)

put $x = \pi/4$; $a + \frac{C}{2} = 0$ --- (iii)

adding (i) and (ii)

$2a + C = 0 \Rightarrow C = -2a$

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put in (iii)

$$a + \frac{(-2a)}{2} = 0$$

$$a - a = 0$$

There exists infinite many solution so, 1, $\cos 2x$, $\cos^2 x$ are linearly independent.

Q4

Solve the differential equation subject to initial condition

$$y''' + 2y'' - 5y' - 6y = 0$$

$$y(0) = y'(0) = 0, \quad y''(0) = 1$$

soln:

$$(D^3 + 2D^2 - 5D \cancel{+} -6)y = 0$$

Auxilary eq is

$$m^3 + 2m^2 - 5m - 6 = 0$$

solve by using synthetic division

$$\begin{array}{r|rrrr} -1 & 1 & 2 & -5 & -6 \\ & \downarrow & -1 & -1 & 6 \\ \hline & 1 & 1 & -6 & \boxed{0} \end{array}$$

$m = -1$ are one root

Depressed eq is

$$m^2 + m - 6 = 0$$

$$m^2 + 3m - 2m - 6 = 0$$

$m(m+3)-2(m+3)=0$

$(m+3)(m-2)=0$

$m = 2, -3$

$y_c = C_1e^{-x} + C_2e^{2x} + C_3e^{-3x}$

For condition

$y(0) = 0$

$0 = C_1 + C_2 + C_3$

$C_1 + C_2 + C_3 = 0$ --- (i)

$y' = -C_1e^{-x} + 2C_2e^{2x} - 3C_3e^{-3x}$

For condition

$y'(0) = 0$

$0 = -C_1 + 2C_2 - 3C_3$

$2C_2 - C_1 - 3C_3 = 0$ --- (ii)

$y'' = C_1e^{-x} + 4C_2e^{2x} + 9C_3e^{-3x}$

For condition

$y''(0) = 1$

$1 = C_1 + 4C_2 + 9C_3$

$C_1 + 4C_2 + 9C_3 = 1$ --- (iii)

Adding eq (i) and (ii)

$3C_2 - 2C_3 = 0$ --- (iv)

$\cancel{C_2 = \frac{2C_3}{3}}$

Adding eq (ii) and (iii)

$6C_2 + 6C_3 = 1$ --- (v)

Multiply 3 by eq (iv) and add in

$$9C_2 - 6C_3 = 0$$
$$\underline{6C_2 + 6C_3 = 1}$$
$$15C_2 = 1$$
$$C_2 = 1/15$$

put in (iv)

$$3\left(\frac{1}{15}\right) - 2C_3 = 0$$

$$2C_3 = 1/5$$
$$C_3 = 1/10$$

put values in eq (i)

$$C_1 + \frac{1}{15} + \frac{1}{10} = 0$$

$$C_1 = \frac{-10-15}{150} = \frac{-25}{150} = -\frac{1}{6}$$

$$y = -\frac{1}{6}e^{-x} + \frac{1}{15}e^{2x} + \frac{1}{10}e^{-3x}$$ **Ans:**